

Shuja Chaudhry

Hardware Design Engineer, PCB and Embedded Systems

Selected boards from over 100 delivered PCB projects, taken from requirements and component selection through schematic capture, multilayer layout, fabrication files and hardware bring up. The focus of this collection is battery powered, wireless and sensor based electronics: compact BLE devices, power management chains, analog front ends and safety critical drivers.

16

projects in this
portfolio

100+

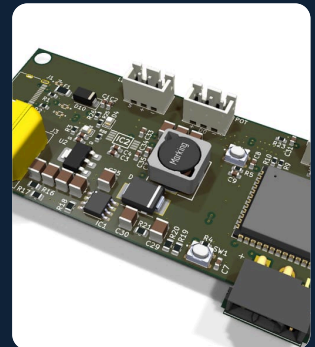
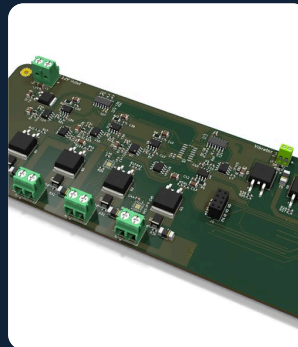
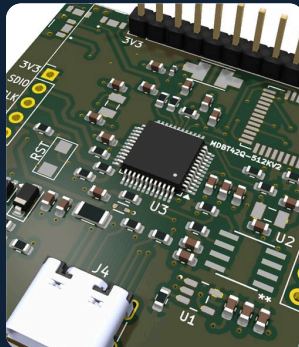
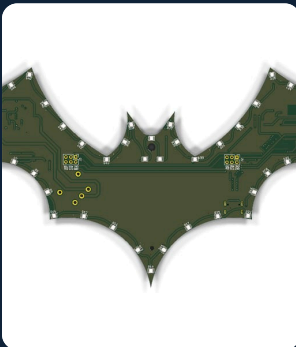
PCB projects
delivered

2 to 4

layer designs
shown

KiCad

and Altium
Designer



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Portfolio
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Skills and Workflow

PCB DESIGN

- Multilayer schematic capture and PCB layout, 2 to 4 layers shown here
- KiCad and Altium Designer
- Custom outlines, castellated modules, HAT and carrier form factors
- DRC clean layouts within JLCPCB and PCBWay assembly capability

EMBEDDED PLATFORMS

- ESP32, ESP32-S3, ESP32-C3, STM32G0, nRF52832, RP2040
- Bluetooth Low Energy, WiFi, LoRa, Zigbee, 433 MHz, Ethernet with PoE
- UART, SPI, I2C, I2S, CAN, USB

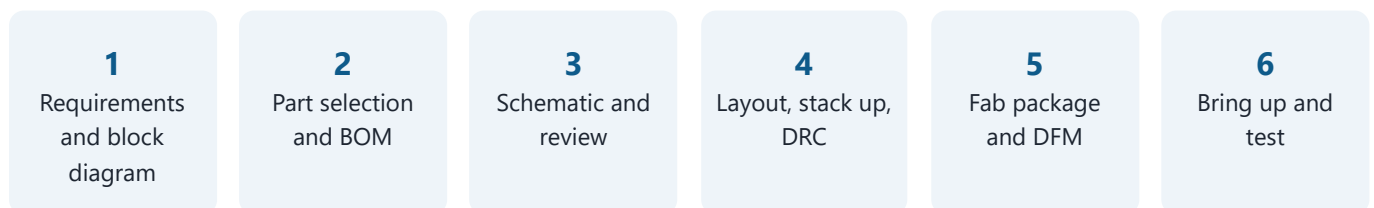
POWER AND ANALOG

- Single cell Li ion and LiPo charging, protection and power path design
- Buck, boost and buck boost converters, LDOs, power multiplexing
- Load cell, current sense, temperature, motion and gas sensor front ends
- ESD, TVS, polyfuse and reverse polarity protection

DELIVERABLES

- Schematics with design notes
- Gerbers, drill files and stack up notes
- BOM with manufacturer part numbers and stock check
- Pick and place files and assembly drawings
- Bring up plan, test points and design review notes

HOW I WORK



Every board in this portfolio was delivered as a manufacturing ready package. Where photographs are shown, the board was fabricated, assembled and powered up.

Project Index

No.	Project	Platform	Board	Focus
1	ESP32 BLE Wearable Badge with 40 RGB LEDs	ESP32 with BLE	Custom outline	Wearable, BLE
2	STM32 plus nRF52832 Secure BLE Device	STM32G0B1 and nRF52832 (Raytac MDBT42Q)	4 layer	BLE, STM32
3	PEMF Therapy Controller HAT for Raspberry Pi 5	Raspberry Pi 5 HAT, TLC555 timing, TC4427 drivers	2 layer	Therapy device, Coil driver
4	BLE Motion Sensor Module with Castellated Edges	BLE SoC with onboard antenna	Castellated module	BLE, Motion sensing
5	ESP32-S3 Portable Audio Player with Battery Management	ESP32-S3-WROOM-1 N16R8	4 layer	Battery management, 4 layer
6	ESP32-C3 Load Cell Sensor Board with HX711	ESP32-C3-MINI	2 layer	Analog front end, Sensor
7	ESP32 PoE IoT Gateway	ESP32 with W5500 Ethernet and Zigbee radio	Compact multi sensor gateway	PoE, Ethernet
8	LoRa and BLE Handheld Remote Controller	LoRa module and HM-11 BLE module	2 layer	LoRa, BLE
9	ESP32-S3 Exoskeleton Controller with CAN Bus	ESP32-S3-WROOM-1	4 layer	CAN bus, 4 layer
10	ESP32-S3 Air Quality Sensor Node with Ethernet and PoE	ESP32-S3	2 layer	Sensors, PoE
11	Raspberry Pi Pico W Sensor Carrier with GPS and IMU	Raspberry Pi Pico W (RP2040)	2 layer	RP2040, GPS
12	Battery Powered Environmental Monitoring Node	ESP32 development module	2 layer carrier	Battery powered, Sensors
13	ESP32-S3 Smart AC Controller with IR and Climate Sensing	ESP32-S3-WROOM-1	2 layer	WiFi, IR
14	Vehicle LED Lighting Controller	MP9929 buck controller, MOSFET outputs	4 layer	Automotive, 4 layer
15	Isolated Relay and Servo Boards for a Launcher Platform	CPC1706Y relays, TLP240A inputs, servo drivers	2 layer	Solid state relays, Servo outputs
16	12 V Industrial Tap Detector Board	Socketed control module	Industrial sensing board	Industrial, 12 V

ESP32 BLE Wearable Badge with 40 RGB LEDs

Custom outline, LiPo powered, assembled and tested

Wearable

BLE

Battery powered

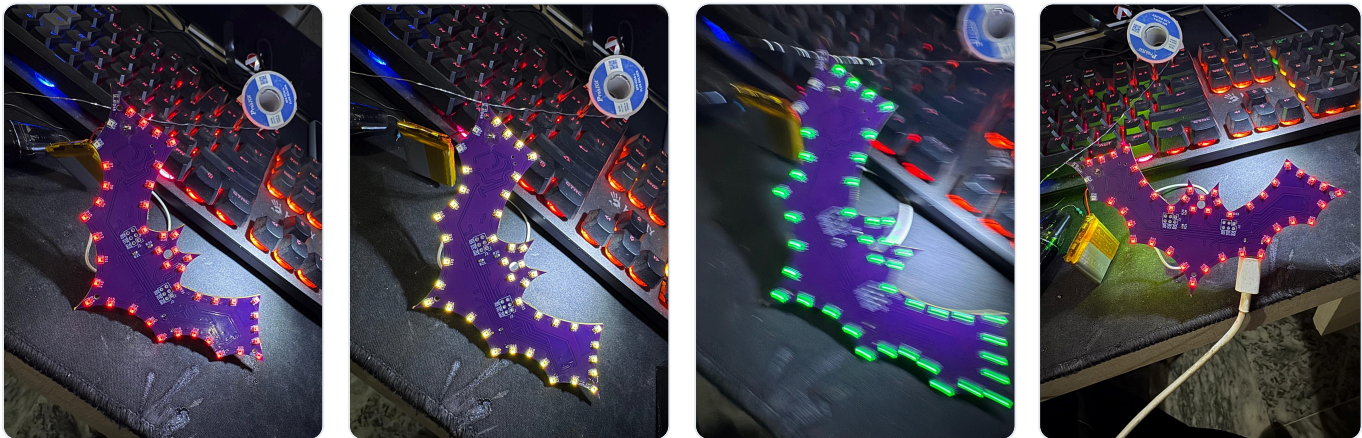
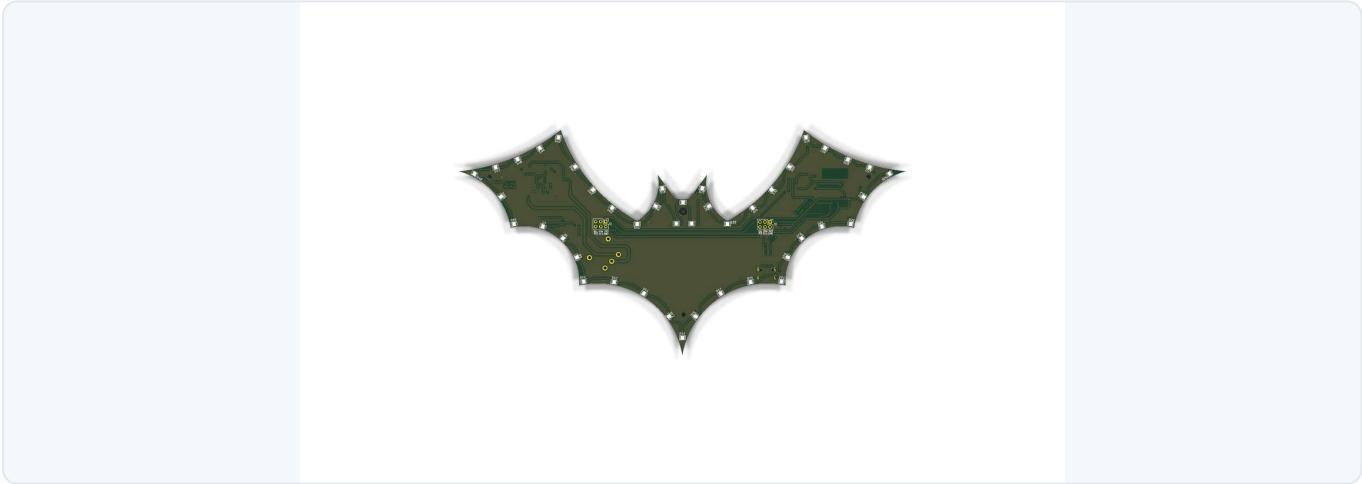
Assembled and tested

Custom outline PCB for a wearable ESP32 badge with BLE and 40 addressable RGB LEDs, running from USB-C or an onboard LiPo cell. The hard part was power: a buck boost converter holds the 5 V LED rail steady as the battery drops below 5 V, the ESP32 runs from its own 3.3 V rail so LED switching current never reaches the MCU, and power path switching hands over between USB and battery without a glitch or a reset. Part choice and layout were kept inside JLCPCB assembly capability so the board could be built without redesign. The photos show the assembled board running colour animations from the battery.

Platform	ESP32 with BLE
Board	Custom outline, JLCPCB assembly ready
Interfaces	USB-C, addressable LED chain, I2C header
Power	USB-C or single cell LiPo, buck boost 5 V rail, 3.3 V MCU rail
Delivered	Schematic, layout, Gerbers, BOM, assembled prototype

KEY DESIGN POINTS

- Buck boost converter keeps the LED rail stable across the full LiPo discharge curve
- Separate 3.3 V MCU rail isolates the radio and processor from LED switching noise
- Seamless USB to battery power path handover with no brownout or reset



Assembled hardware photographs

STM32 plus nRF52832 Secure BLE Device

Security

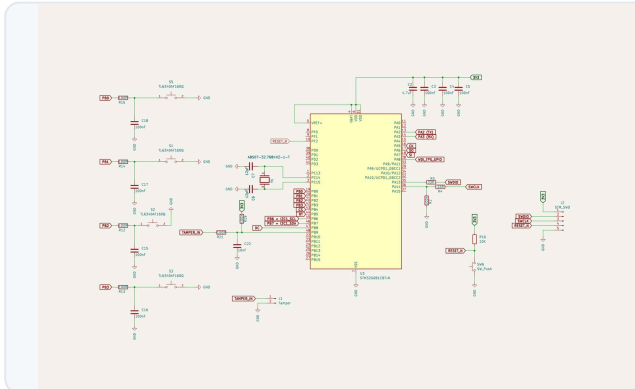
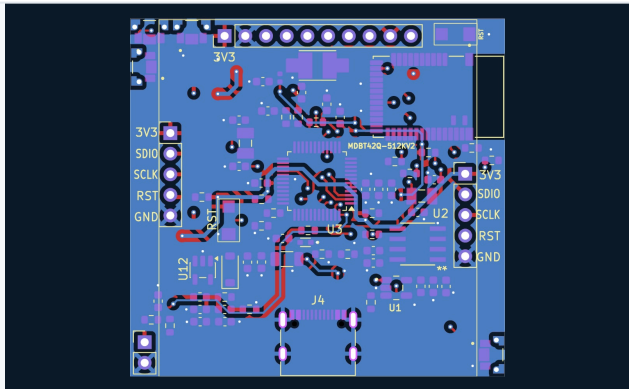
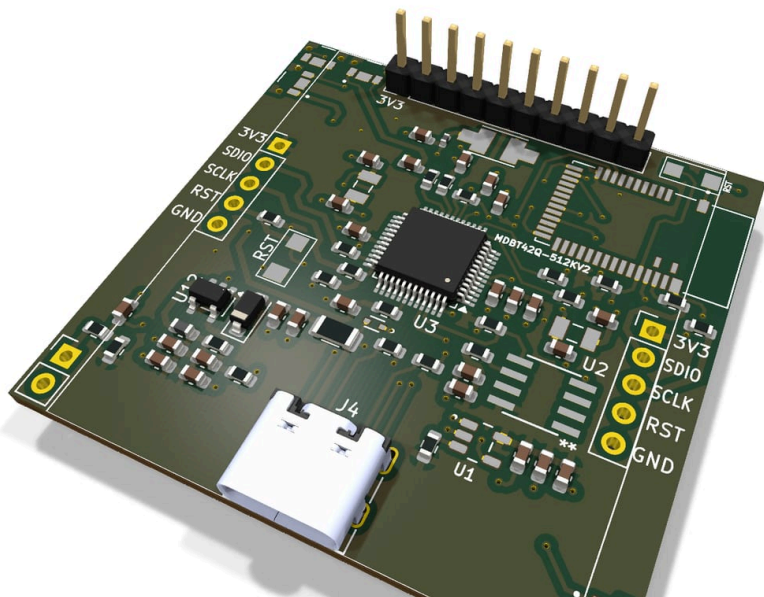
Four layer, dual MCU, secure element and tamper detection

Four layer KiCad PCB design for a dual MCU secure BLE product. An STM32G0B1 handles the application and a Raytac MDBT42Q nRF52832 module handles Bluetooth, with an ATECC608A secure element for key storage and a tamper switch. SY8089A buck from USB-C to 3.3 V, TPS3823 supervisor for clean resets, 32.768 kHz crystal, TFT LCD connector, and separate SWD headers for both processors. 72 components, USB-C with ESD protection and a polyfuse.

KEY DESIGN POINTS

- Certified BLE module keeps the RF section out of the critical path for approvals
- Hardware secure element and tamper input for key storage and intrusion detection
- Supervisor and low noise regulation for reliable resets on a small battery style rail

Platform	STM32G0B1 and nRF52832 (Raytac MDBT42Q)
Board	4 layer, 72 components
Tool	KiCad
Interfaces	BLE, SWD x2, TFT LCD connector, USB-C
Power	USB-C input, SY8089A buck to 3.3 V, TPS3823 supervisor, polyfuse and ESD
Delivered	Schematic, layout, Gerbers, BOM



3D render, routed layout and schematic

PEMF Therapy Controller HAT for Raspberry Pi 5

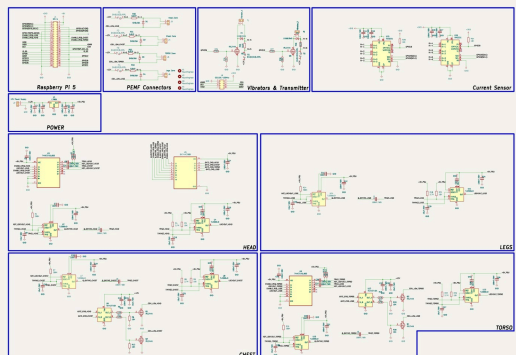
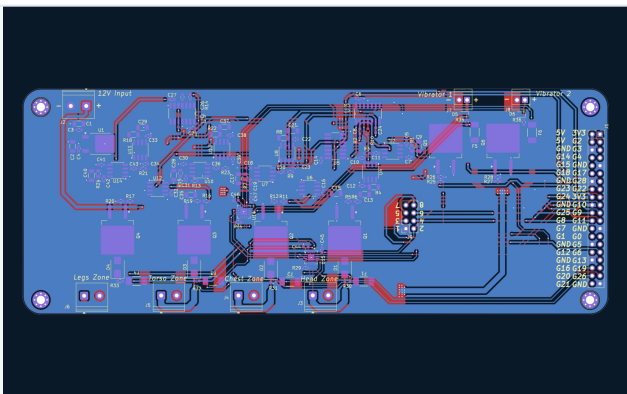
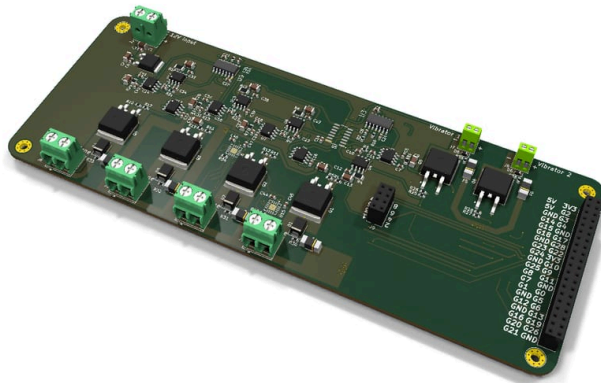
Four zone pulsed electromagnetic field driver with hardware safety lockout

Two layer KiCad PCB design for a Raspberry Pi 5 HAT that drives four pulsed electromagnetic field coils, one each for head, chest, torso and legs. Each zone has a TLC555 timing stage, TC4427 gate driver and IRFS4127 power MOSFET with hardware lockout and enable request lines so the Pi cannot leave a coil on. INA3221 three channel current monitoring, two vibration motor drivers, a CC1101 433 MHz transceiver, L7805 regulator from a 12 V supply, 74HCT logic and TVS protection. 130 components.

KEY DESIGN POINTS

- Hardware interlock per coil so software faults cannot leave a therapy output energised
- Per channel current monitoring for closed loop control and fault reporting
- Vibration motor drivers and a sub GHz radio on the same board for a complete treatment controller

Platform	Raspberry Pi 5 HAT, TLC555 timing, TC4427 drivers
Board	2 layer, 130 components
Tool	KiCad
Interfaces	Pi 40 pin header, I2C (INA3221), SPI (CC1101), coil outputs x4
Power	12 V input, L7805 5 V rail, TVS protection
Delivered	Schematic, layout, Gerbers, BOM



3D render, routed layout and schematic

BLE Motion Sensor Module with Castellated Edges

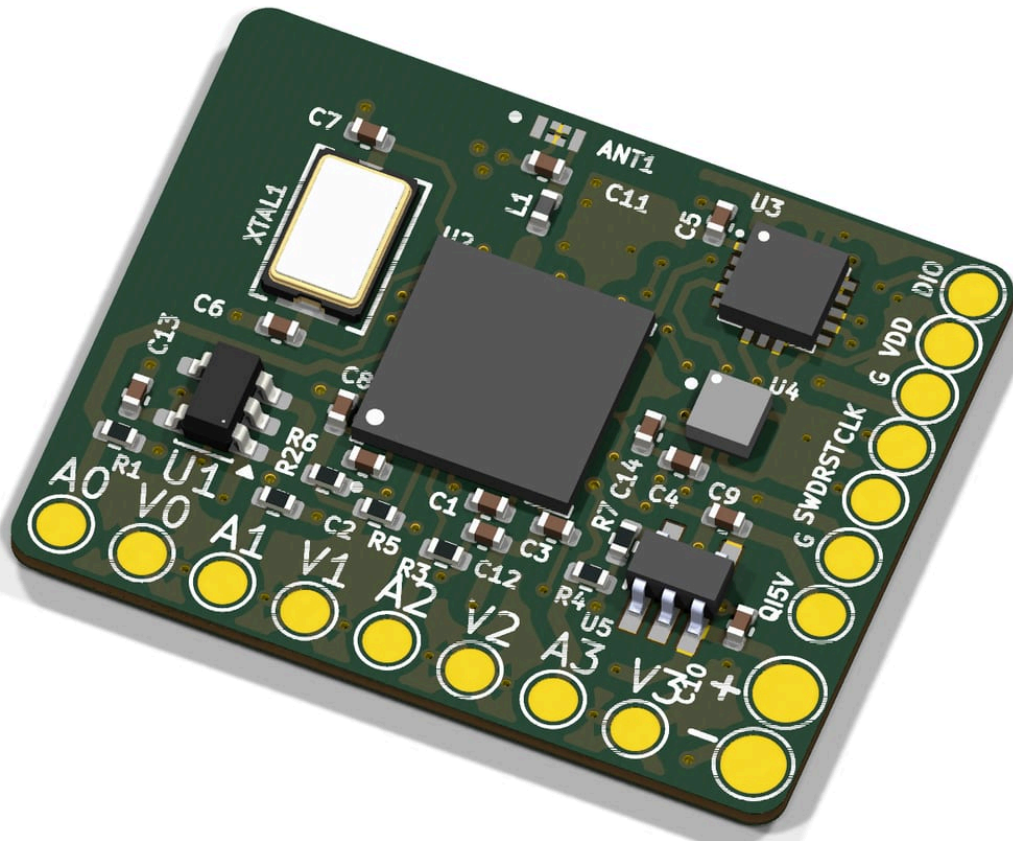
Compact reflow ready module for a smart ball product

Compact castellated PCB module for a smart ball product: BLE SoC with crystal and antenna, motion sensing ICs, SWD debug pads and a single row of castellated pins so the module can be soldered directly onto a carrier board. Antenna keep out, ground plane and battery rails were planned together because in a module this small each one constrains the others. Designed for reflow assembly.

KEY DESIGN POINTS

- Antenna keep out and ground plane planned together on a very small outline
- Castellated edge so the same module drops onto different carrier boards
- Motion sensing front end laid out for low noise on a battery rail

Platform	BLE SoC with onboard antenna
Board	Castellated module, reflow assembly
Interfaces	SWD debug pads, castellated I/O row
Power	Battery rail with local regulation
Delivered	Schematic, layout, Gerbers, BOM



3D render

ESP32-S3 Portable Audio Player with Battery Management

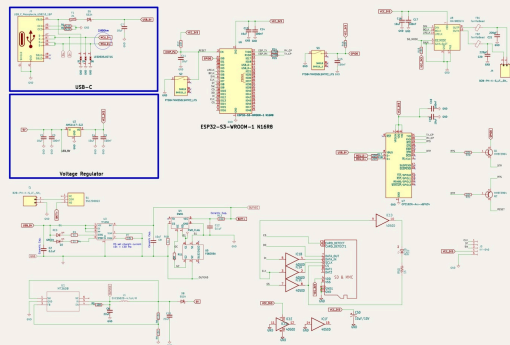
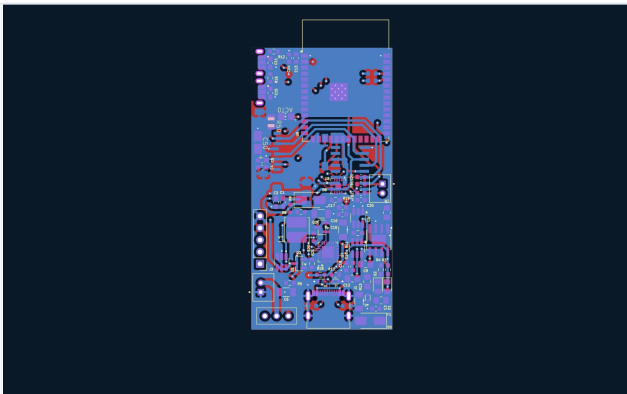
Four layer, full single cell Li ion charge and protection chain

Four layer KiCad PCB design for a battery powered audio device. ESP32-S3-WROOM-1 N16R8 drives a MAX98357A I2S class D amplifier, with a microSD slot for audio files and a CP2102N USB to UART bridge for programming. Full single cell Li ion chain on board: TP4056 charger with the current set resistor documented, DW01 and FS8205A protection, MT3608 boost for the speaker rail and an AMS1117 3.3 V rail, plus a slide power switch and USB-C. 73 components.

KEY DESIGN POINTS

- Complete charge, protect and boost chain for a single Li ion cell
- Amplifier and boost converter placed away from the antenna and analog paths
- Programming bridge and card slot on board for a self contained product

Platform	ESP32-S3-WROOM-1 N16R8
Board	4 layer, 73 components
Tool	KiCad
Interfaces	I2S audio, microSD, USB-C, UART bridge
Power	TP4056 charger, DW01 and FS8205A protection, MT3608 boost, AMS1117 3.3 V
Delivered	Schematic, layout, Gerbers, BOM



3D render, routed layout and schematic

ESP32-C3 Load Cell Sensor Board with HX711

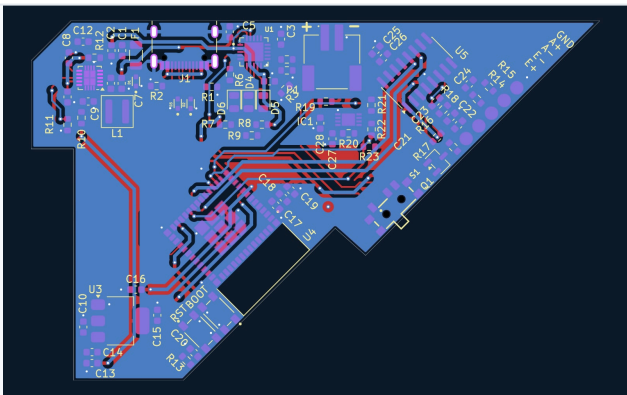
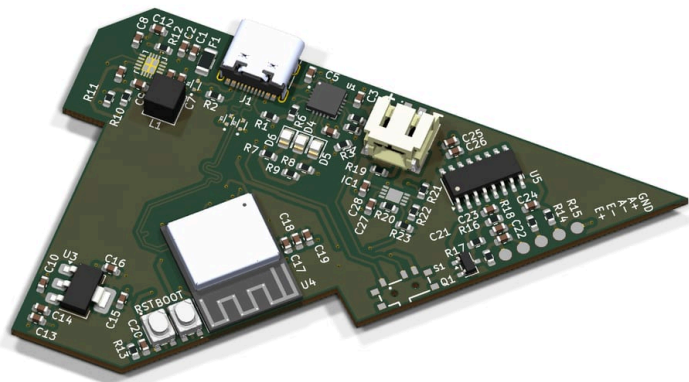
Custom shape, quiet analog front end, wireless weight sensing

Custom shape two layer PCB in KiCad for a wireless weight sensor: ESP32-C3-MINI module, HX711 24 bit load cell amplifier, USB-C power and programming, ESD protection on the USB lines and a connector for the load cell. The analog front end sits on its own quiet area of the board, away from the switching regulator and the antenna keep out. 72 components on a custom outline.

KEY DESIGN POINTS

- 24 bit bridge amplifier placed in a dedicated low noise region
- Regulator and antenna keep out kept clear of the measurement path
- ESD protected USB-C for both power and programming

Platform	ESP32-C3-MINI
Board	2 layer, custom outline, 72 components
Tool	KiCad
Interfaces	HX711 bridge input, USB-C, UART
Power	USB-C 5 V, 3.3 V regulator
Delivered	Schematic, layout, Gerbers, BOM



ESP32 PoE IoT Gateway

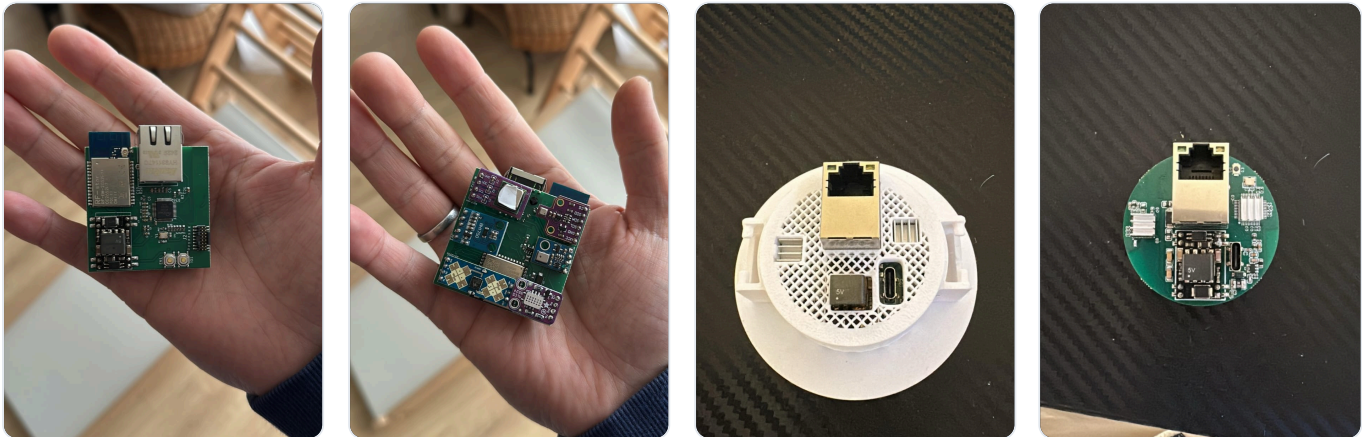
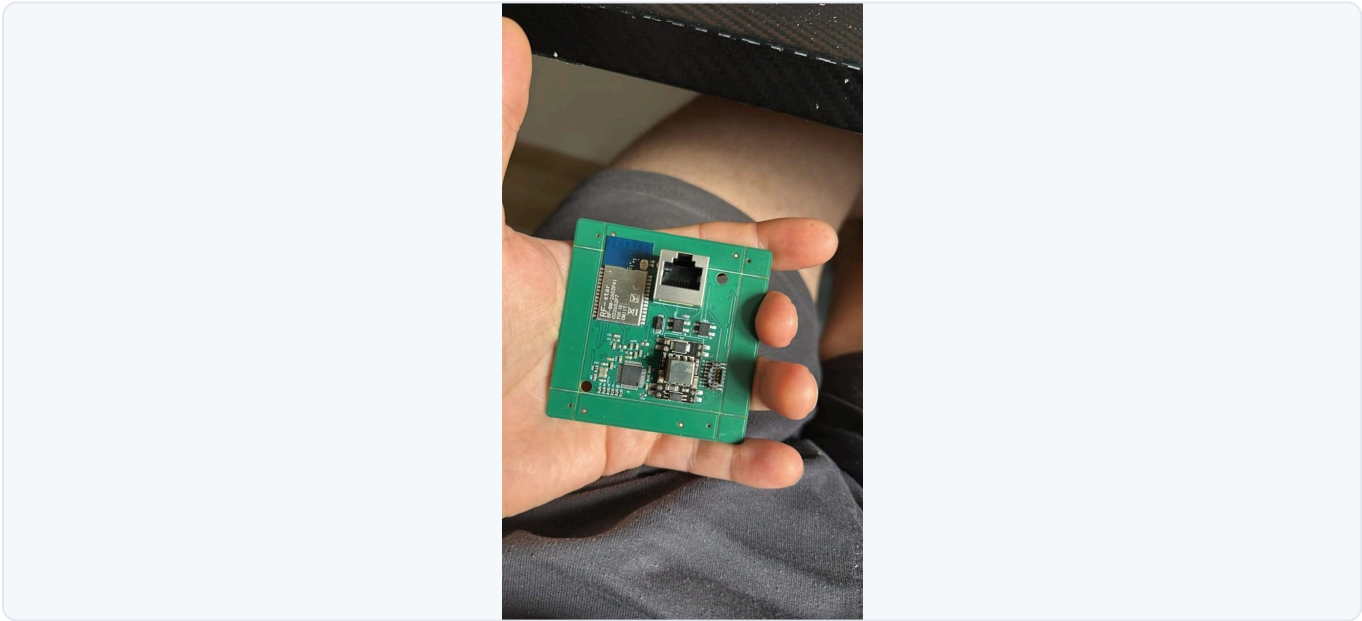
Ethernet powered multi sensor gateway, fabricated, assembled and tested

Custom PCB design for an ESP32 IoT gateway powered over Ethernet. One cable carries power and data, so the PoE front end, the W5500 Ethernet controller, a Zigbee radio and a multi sensor package (CO2, gas, temperature, humidity, light, pressure) all share one compact board. The switching supply was floorplanned away from the radio, and the sensors were placed by distance from self heating parts, because a temperature reading is only as good as its placement. The photos show the board as returned from manufacturing and powered up.

KEY DESIGN POINTS

- PoE front end and Ethernet controller integrated with a radio on one compact board
- Thermal aware sensor placement for accurate temperature readings
- Full fabrication package delivered and verified on real hardware

Platform	ESP32 with W5500 Ethernet and Zigbee radio
Board	Compact multi sensor gateway
Interfaces	Ethernet with PoE, Zigbee, I2C sensors, UART
Power	PoE input, isolated switching supply, 3.3 V rail
Delivered	Schematic, layout, Gerbers, BOM, pick and place, tested prototype



Assembled hardware photographs

LoRa and BLE Handheld Remote Controller

USB-C charging

Production batch

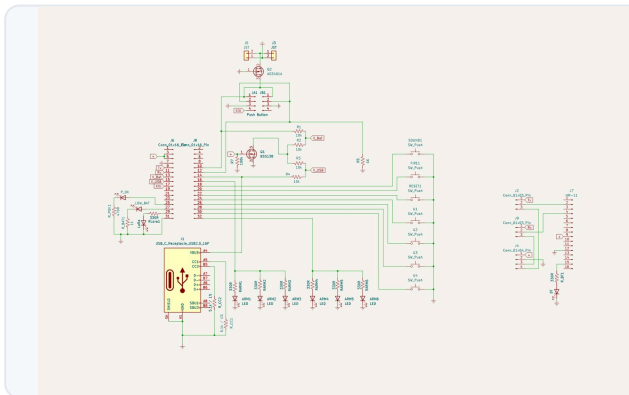
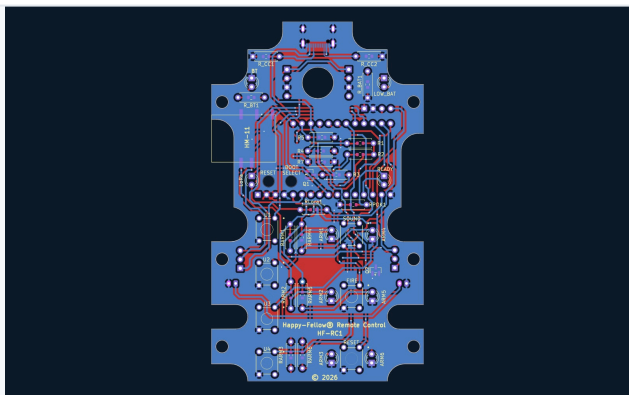
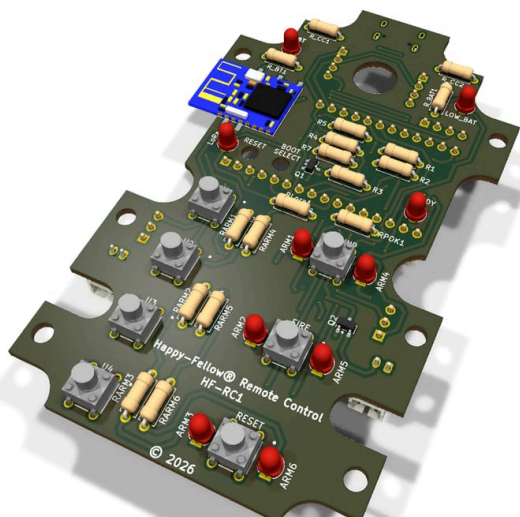
Two layer, custom hand grip outline, built as a small production batch

Two layer KiCad PCB design for the HF-RC1 handheld remote of the Happy Fellow dog ball launcher platform. A LoRa module for long range control and an HM-11 BLE module for phone pairing share one board with six tactile buttons (arm, fire and mode), status LEDs for power, low battery and Bluetooth, USB-C charging and a JST battery connector. A P channel load switch and BSS138 level shifters keep the radios on their own rails. 67 components on a custom outline shaped to the hand grip.

KEY DESIGN POINTS

- Two radios on one board with independent switched rails
- Outline shaped to the enclosure and hand grip
- Delivered and manufactured as a small production run

Platform	LoRa module and HM-11 BLE module
Board	2 layer, custom outline, 67 components
Tool	KiCad
Interfaces	LoRa, BLE, UART, six tactile inputs, status LEDs
Power	USB-C charging, JST battery input, P channel load switch
Delivered	Schematic, layout, Gerbers, BOM, production batch



3D render, routed layout and schematic

ESP32-S3 Exoskeleton Controller with CAN Bus

Power management

Medical robotics

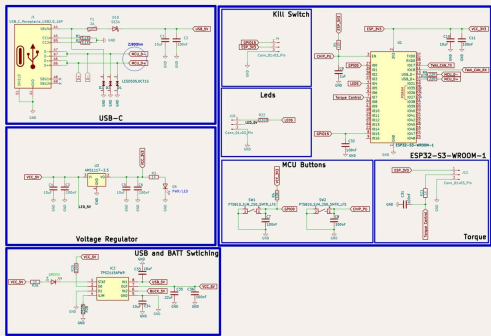
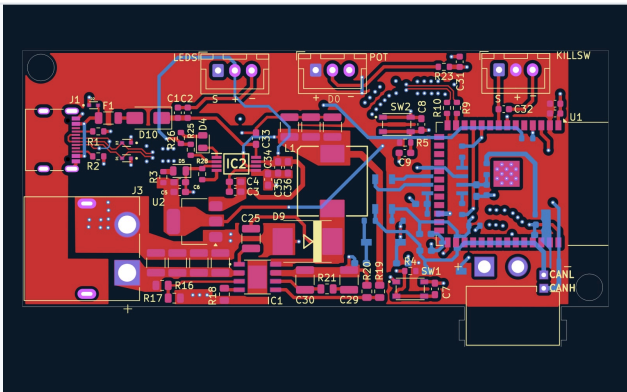
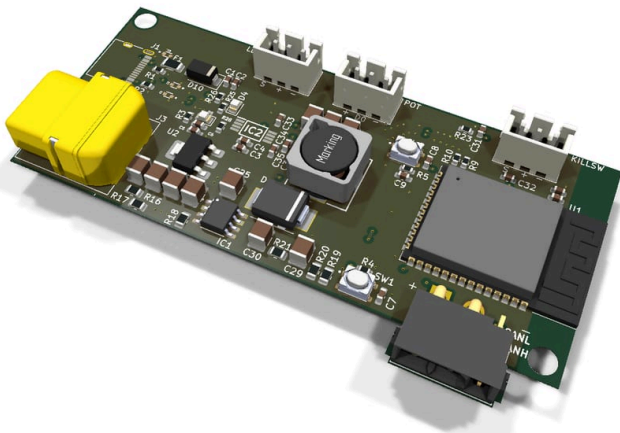
Four layer, 60 V input, USB and battery power multiplexing

Four layer KiCad PCB design for the control board of a powered exoskeleton. ESP32-S3-WROOM-1 with a TPS54560 60 V synchronous buck taking the battery pack down to 5 V, an AMS1117 3.3 V rail and a TPS2115A power multiplexer that switches between USB-C and battery without a reset. CAN bus through a TJA1051 transceiver with split termination and PESD2CANFD surge protection, XT60 and XT30 power connectors, a hardware kill switch, polyfuse and ESD protection on USB. 88 components.

KEY DESIGN POINTS

- High voltage synchronous buck with tight loop layout on a 4 layer stack
- Glitch free power multiplexing between USB and the battery pack
- Protected CAN interface and hardware kill switch for a safety critical actuator system

Platform	ESP32-S3-WROOM-1
Board	4 layer, 88 components
Tool	KiCad
Interfaces	CAN (TJA1051), USB-C, XT60 and XT30 power
Power	TPS54560 60 V buck, AMS1117 3.3 V, TPS2115A power mux, polyfuse, ESD
Delivered	Schematic, layout, Gerbers, BOM, pick and place



3D render, routed layout and schematic

ESP32-S3 Air Quality Sensor Node with Ethernet and PoE

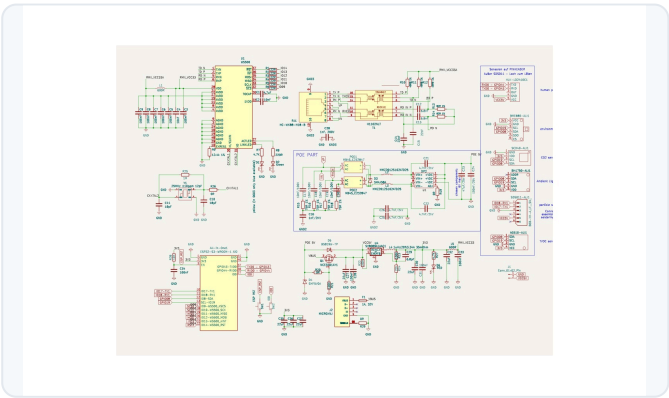
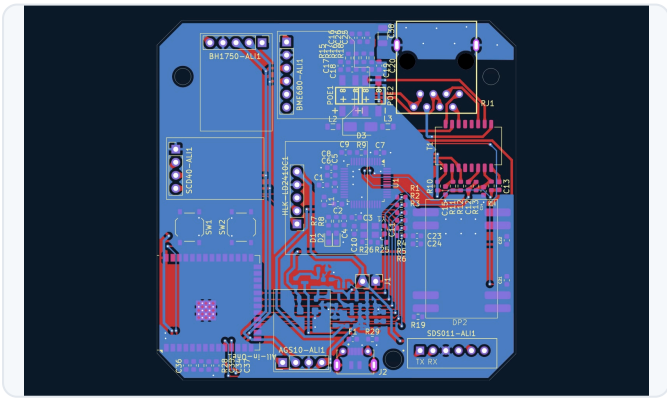
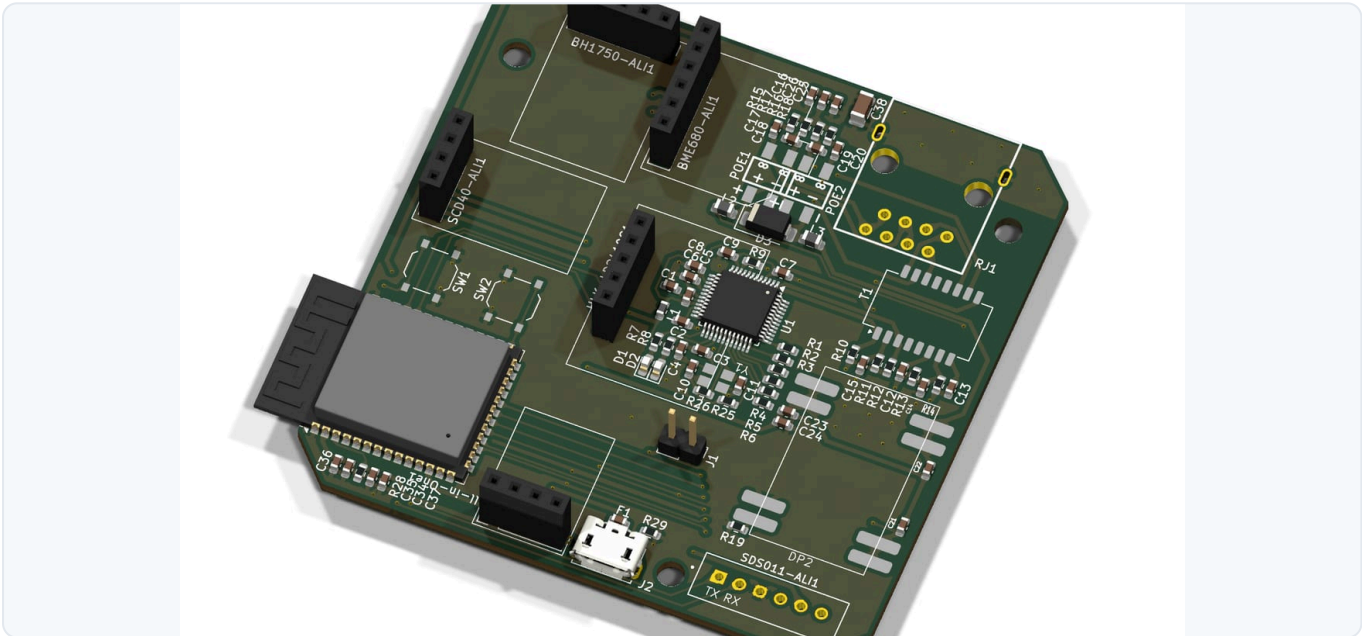
Two layer, 99 components, seven sensors on one compact outline

Two layer PCB design in KiCad for an all in one indoor air quality node: ESP32-S3, SCD40 CO2 sensor, BME680 gas, temperature, humidity and pressure, BH1750 light, SDS011 particulate sensor and a 24 GHz presence radar, with W5500 Ethernet and PoE power on the same board. 99 components on a compact outline with an RJ45 magjack, USB-C for programming and pin headers for the sensor modules. DRC clean layout.

KEY DESIGN POINTS

- Seven sensing functions integrated on a compact two layer board
- Ethernet, PoE and WiFi on the same design without interference
- Module headers so sensors can be swapped during evaluation

Platform	ESP32-S3
Board	2 layer, 99 components
Tool	KiCad
Interfaces	Ethernet with PoE, I2C sensors, UART (SDS011, radar), USB-C
Power	PoE input, 5 V and 3.3 V rails
Delivered	Schematic, layout, Gerbers, BOM



3D render, routed layout and schematic

Raspberry Pi Pico W Sensor Carrier with GPS and IMU

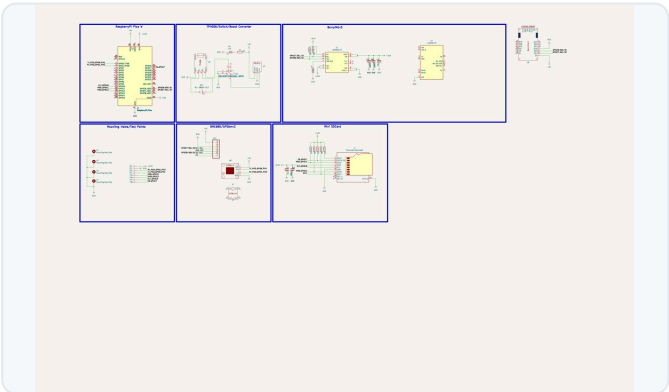
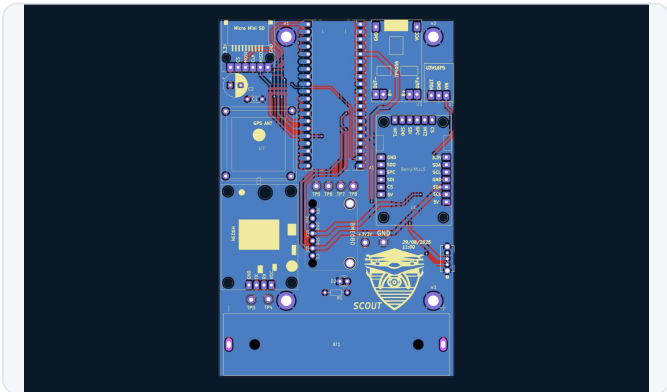
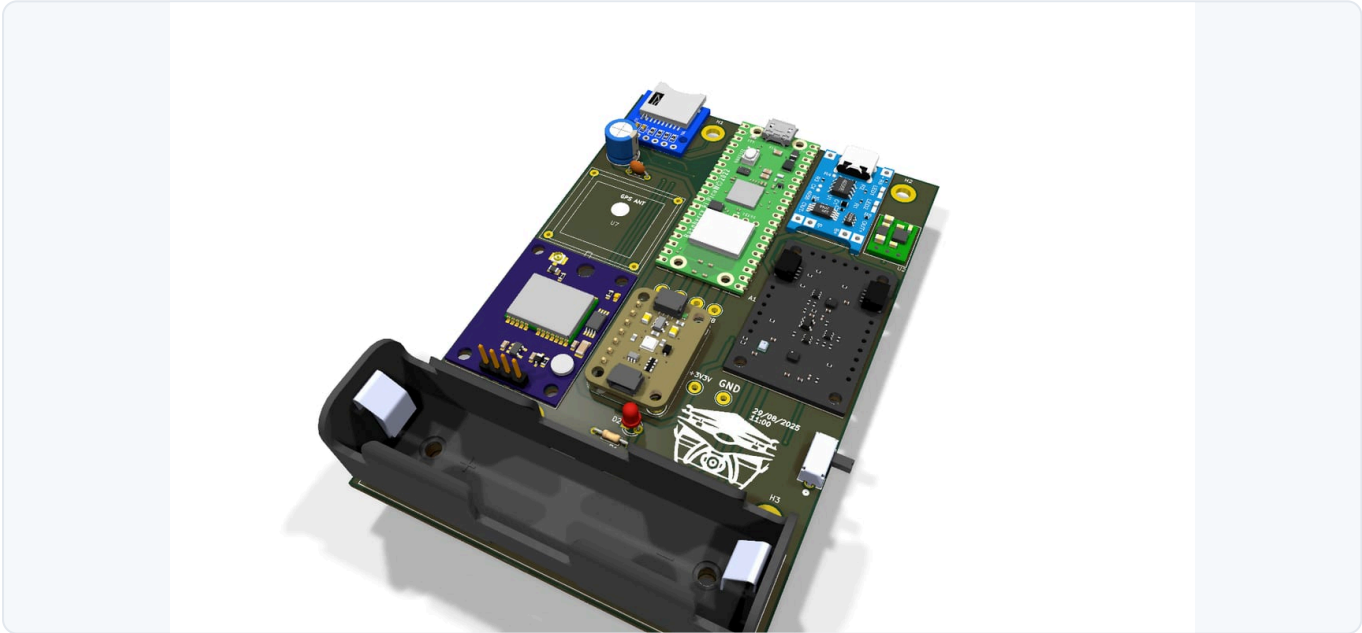
Two layer outdoor tracker carrier with 18650 charging

Two layer KiCad carrier board for a Raspberry Pi Pico W outdoor tracker. Footprints for a NEO-6M GPS module with antenna, a BerryIMU v3 (LSM6DSL accelerometer and gyroscope, LIS3MDL magnetometer), BME680 environmental sensor and a microSD socket with card detect, plus an 18650 cell holder with TP4056 charging, boost converter and power switch. Test points and mounting holes for the enclosure. 31 components, designed for hand assembly.

KEY DESIGN POINTS

- Motion, position and environmental sensing on one carrier
- Single cell charge and boost chain for long outdoor run time
- Test points and enclosure mounting planned from the start

Platform	Raspberry Pi Pico W (RP2040)
Board	2 layer, 31 components
Tool	KiCad
Interfaces	UART GPS, I2C IMU and BME680, SPI microSD
Power	18650 cell, TP4056 charger, boost converter
Delivered	Schematic, layout, Gerbers, BOM



3D render, routed layout and schematic

Battery Powered Environmental Monitoring Node

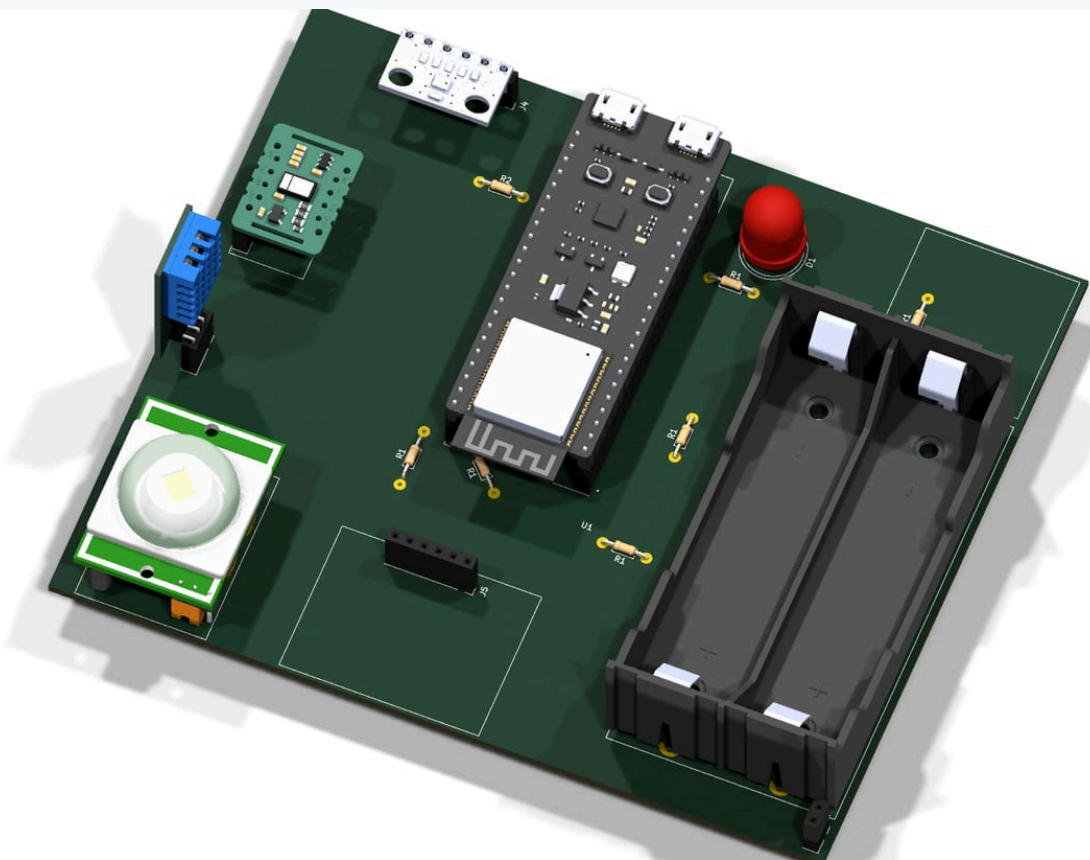
Two layer carrier with dual 18650 cells

Two layer carrier PCB for a battery powered monitoring node: ESP32 development module, dual 18650 cell holders, motion sensor, temperature and humidity sensor, status LED and connectors for external probes. Designed for quick prototype assembly by hand and for a long run time from the two cells.

KEY DESIGN POINTS

- Dual cell holder layout with the sensors kept away from the cells
- Hand assembly friendly footprints for fast prototype turns
- External probe connectors for field measurements

Platform	ESP32 development module
Board	2 layer carrier
Interfaces	I2C sensors, probe connectors, status LED
Power	Dual 18650 cells
Delivered	Schematic, layout, Gerbers, BOM



3D render

ESP32-S3 Smart AC Controller with IR and Climate Sensing

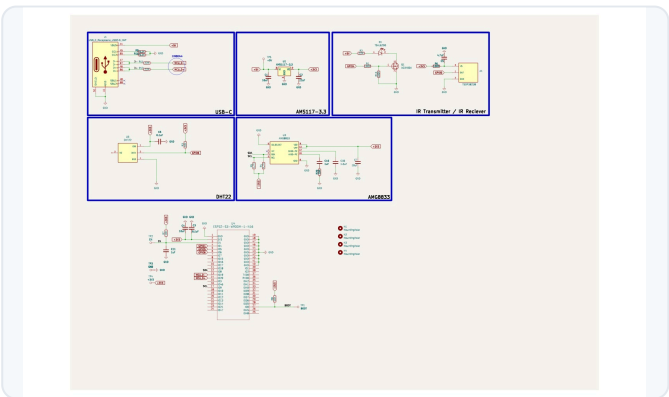
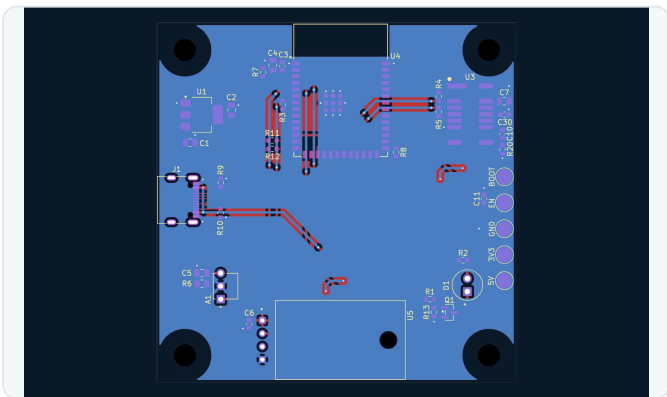
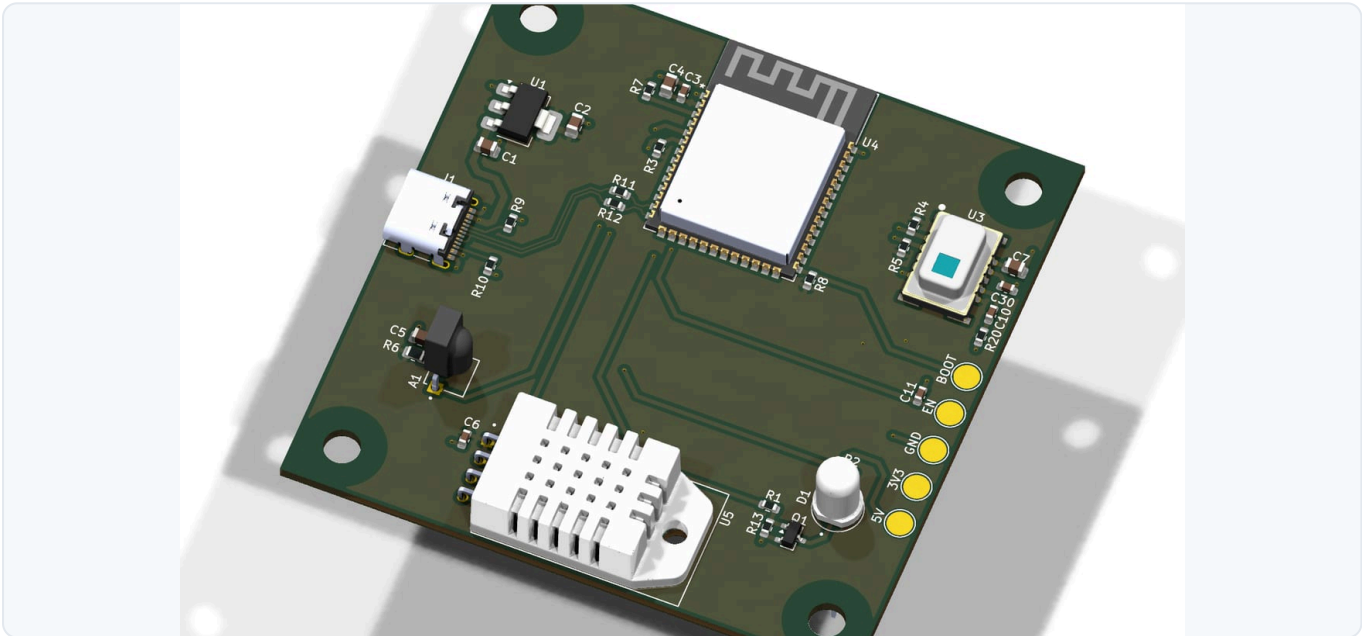
Two layer WiFi controller sized for a small wall enclosure

Two layer KiCad PCB design for a WiFi air conditioner controller built around the ESP32-S3-WROOM-1. USB-C input with an AMS1117 3.3 V rail, IR transmitter and receiver for controlling and learning remote codes, a DHT22 temperature and humidity sensor and an AMG8833 thermal array, with test points and a header for programming. 41 components, four mounting holes, sized for a small wall enclosure.

KEY DESIGN POINTS

- IR transmit and learn path with a thermal camera on a small board
- Programming header and test points for easy bring up
- Ready for JLCPCB fabrication and assembly

Platform	ESP32-S3-WROOM-1
Board	2 layer, 41 components
Tool	KiCad
Interfaces	IR TX and RX, I2C (AMG8833), single wire DHT22, USB-C
Power	USB-C 5 V, AMS1117 3.3 V
Delivered	Schematic, layout, Gerbers, BOM



3D render, routed layout and schematic

Vehicle LED Lighting Controller

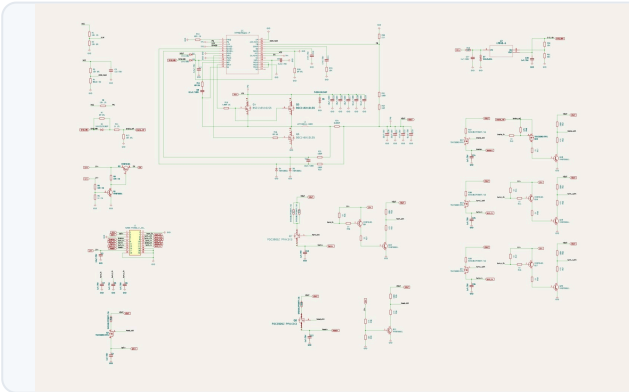
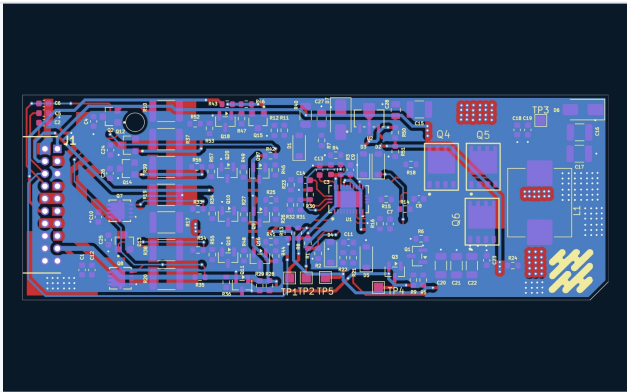
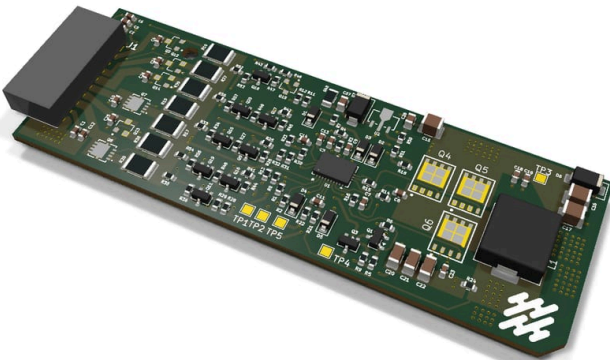
Four layer, 20 A, six high side MOSFET outputs

Four layer KiCad PCB design for an automotive LED lighting control module. Head, tail, brake, left and right turn and horn outputs are switched by BSC146N10LS5 100 V MOSFETs from a 20 pin JST harness connector, with an MP9929 synchronous buck controller for the system rail, an LR8N8 high voltage linear regulator, a 20 A fuse, SMAJ5.0CA transient suppression and Zener clamped gate drives. 121 components with a full stock checked BOM.

KEY DESIGN POINTS

- Copper and via planning for 20 A on a 4 layer stack
- Transient and reverse protection for a vehicle electrical environment
- Stock checked BOM so the board can be assembled straight away

Platform	MP9929 buck controller, MOSFET outputs
Board	4 layer, 121 components
Tool	KiCad
Interfaces	20 pin JST harness, six switched outputs
Power	12 V vehicle supply, 20 A fuse, SMAJ5.0CA TVS
Delivered	Schematic, layout, Gerbers, BOM, pick and place



3D render, routed layout and schematic

Isolated Relay and Servo Boards for a Launcher Platform

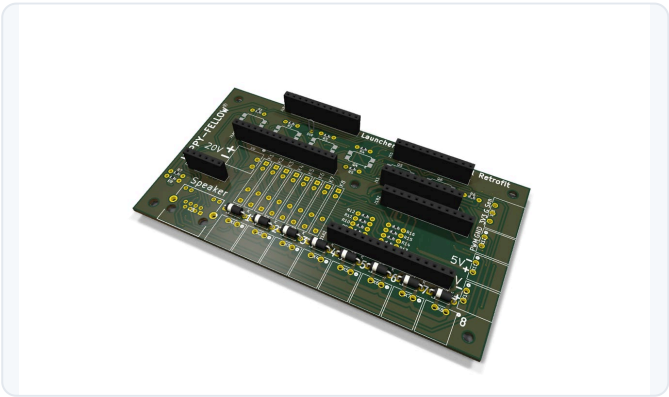
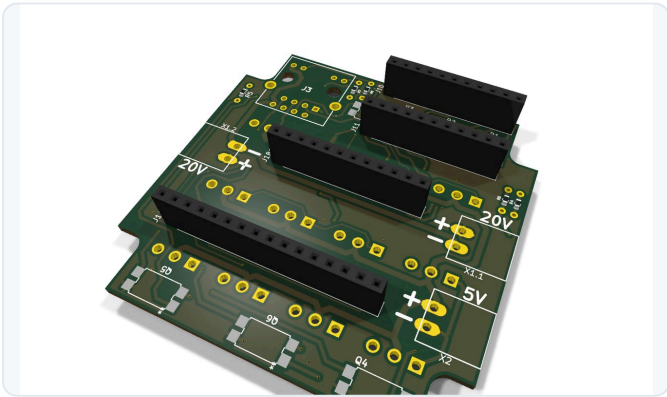
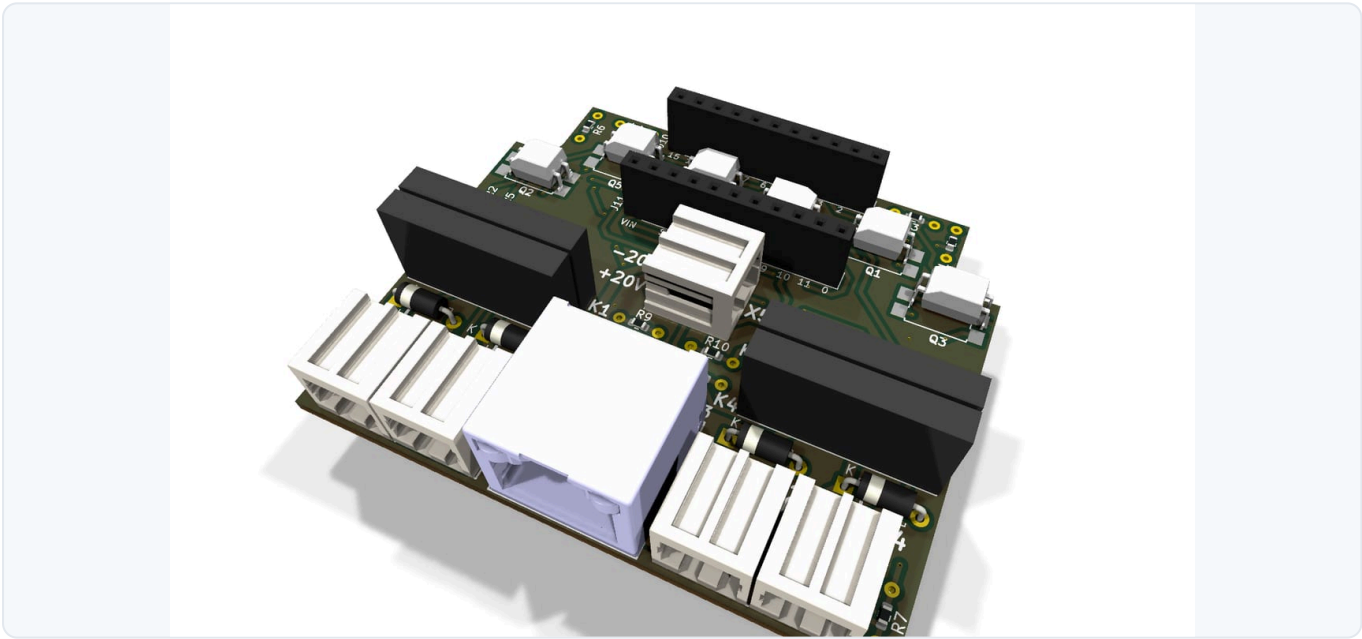
Three two layer boards sharing one controller pinout

Three two layer KiCad boards for the Happy Fellow dog ball launcher. AeroPro4 is a four channel launcher driver with CPC1706Y 4 A solid state relays and TLP240A photo MOSFET inputs. ServoPro12 is a twelve channel servo output board. AeroPro Retrofit combines launcher, speaker and controller headers on one board to upgrade older units. All three use Wago 734 series push in terminals for tool free field wiring and share one connector pinout so they plug into the same controller. 50 to 92 components each.

KEY DESIGN POINTS

- Optically isolated inputs and solid state relay outputs
- One shared pinout across a family of three boards
- Tool free push in terminals for installation in the field

Platform	CPC1706Y relays, TLP240A inputs, servo drivers
Board	2 layer, 50 to 92 components each
Tool	KiCad
Interfaces	Isolated inputs, relay outputs, 12 servo channels, Wago 734 terminals
Power	Isolated field supply, logic supply
Delivered	Schematics, layouts, Gerbers, BOMs



3D render, routed layout and schematic

12 V Industrial Tap Detector Board

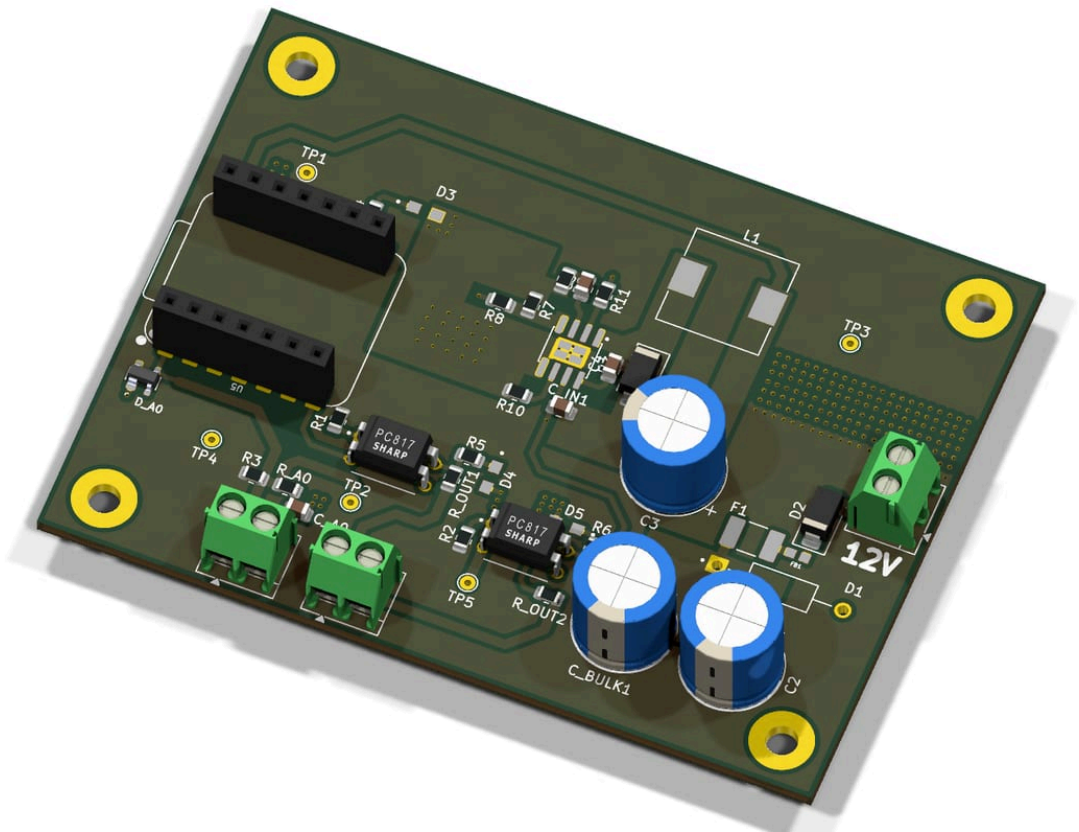
Rugged sensing board with a socketed control module

PCB design for a 12 V industrial sensing board: screw terminal power input with bulk electrolytic filtering, regulated rails, test points, mounting holes and a socketed header for the control module so it can be swapped without desoldering. Laid out for a rugged enclosure with all connections on one edge.

KEY DESIGN POINTS

- Bulk filtering and regulation for a noisy 12 V industrial supply
- Socketed control module for serviceability
- Single edge connections for a rugged enclosure

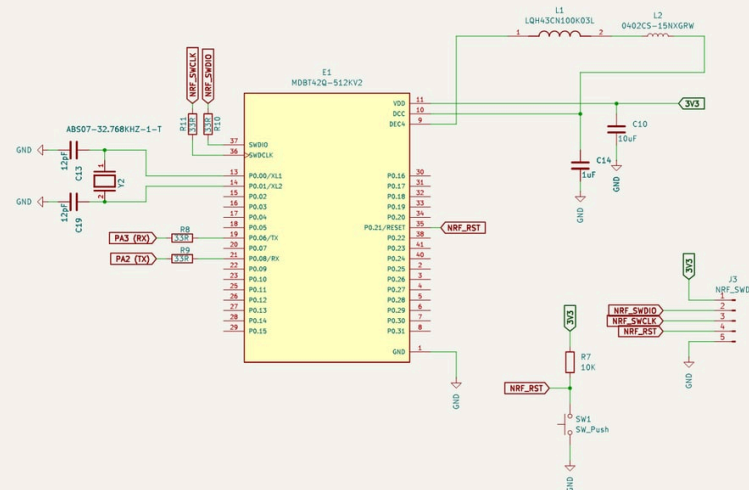
Platform	Socketed control module
Board	Industrial sensing board
Interfaces	Screw terminals, sensor inputs, test points
Power	12 V input, filtered and regulated rails
Delivered	Schematic, layout, Gerbers, BOM



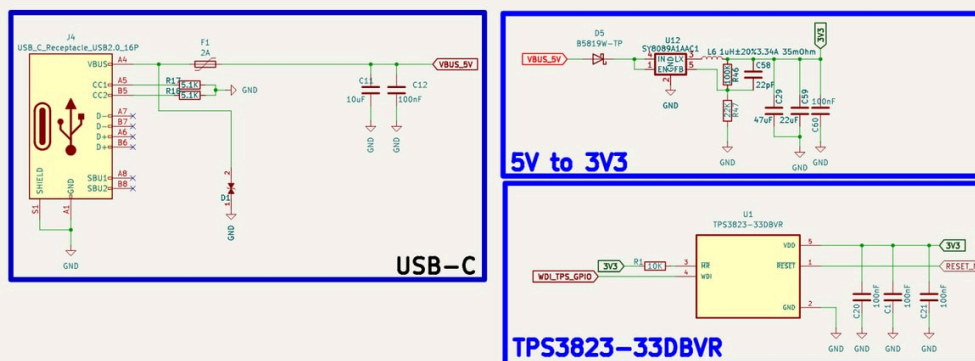
3D render

Appendix: Additional Schematic Sheets and Layouts

Project 02: STM32 plus nRF52832 Secure BLE Device

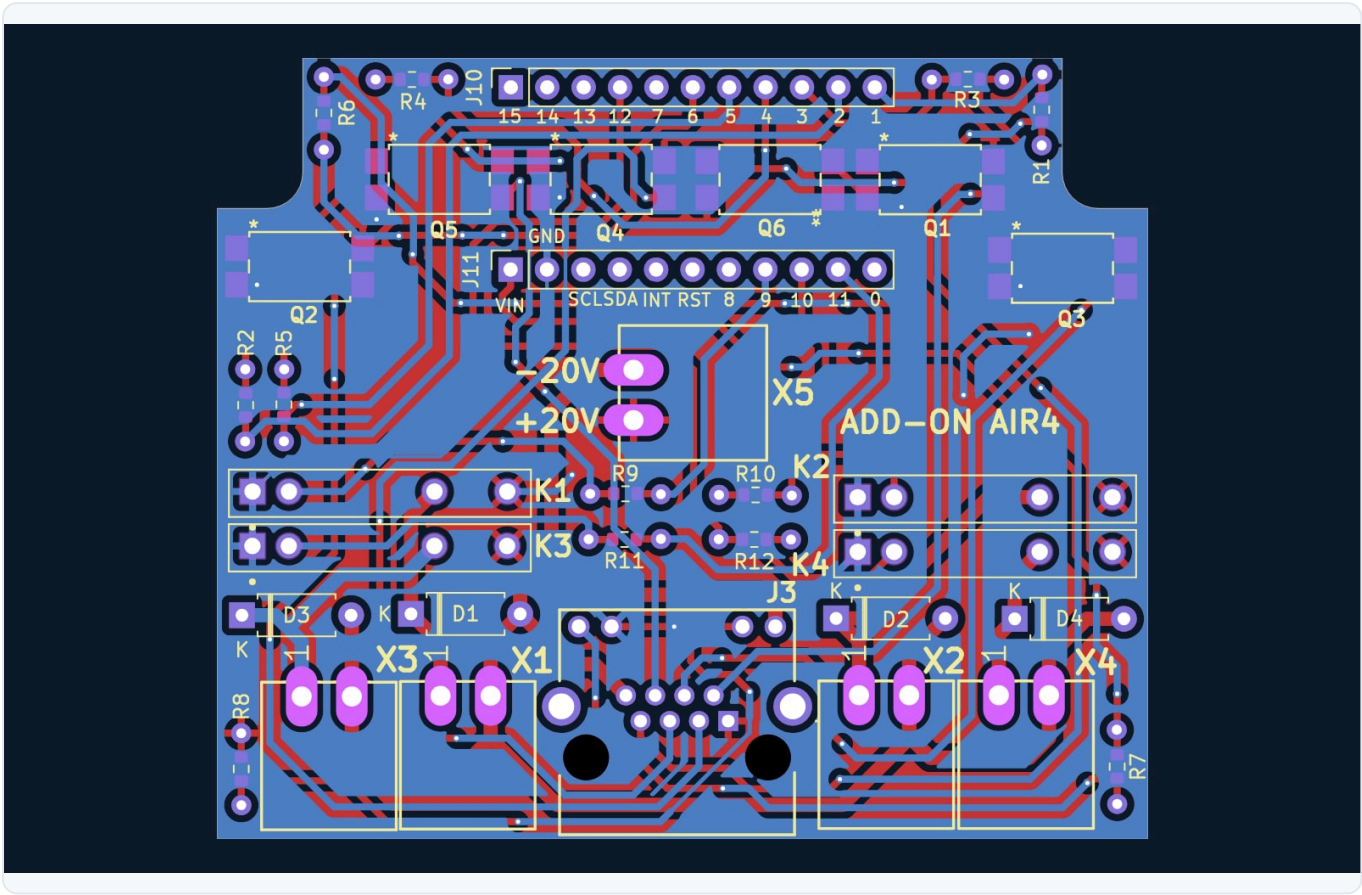


Project 02: STM32 plus nRF52832 Secure BLE Device

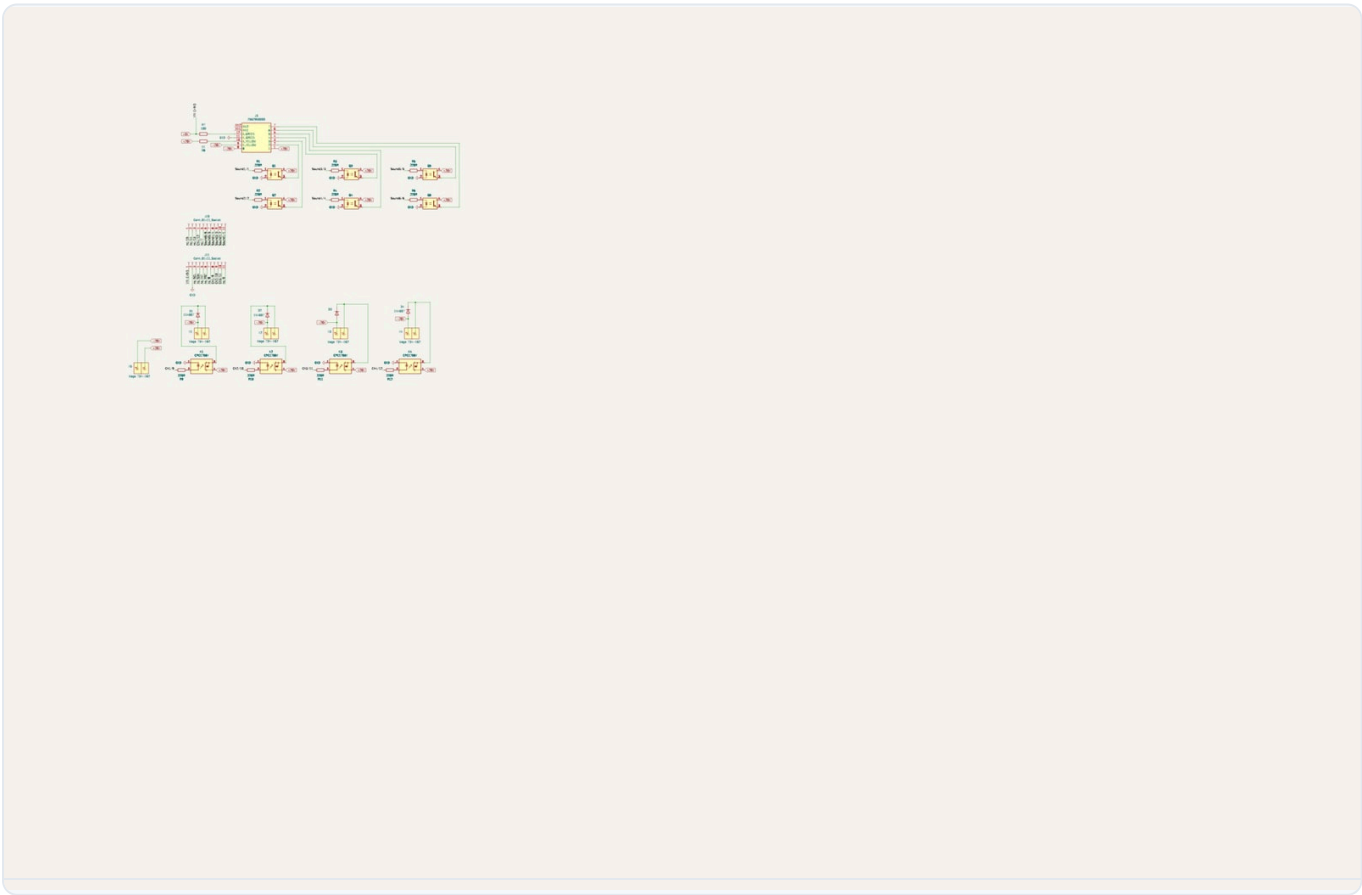


Appendix: Additional Schematic Sheets and Layouts

Project 15: Isolated Relay and Servo Boards for a Launcher Platform



Project 15: Isolated Relay and Servo Boards for a Launcher Platform



Working Together

I am available for remote PCB and embedded hardware work and can align my hours with UK business hours. I am happy to begin with an evaluation task or a scoped trial project so that engineering quality can be judged on real output.

WHAT I CAN TAKE ON

- New board design from a requirements document or block diagram
- Revision of an existing schematic and layout (for example a v2.0 spin)
- Power architecture for battery operated and wearable devices
- Sensor and analog front end design and review
- BLE and wireless module integration with antenna keep out and RF layout
- Fabrication package, DFM review and BOM sourcing
- Bring up plan, test point strategy and debug support

CONTACT

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Native KiCad project files, Gerbers and BOMs for any board shown here can be provided on request, subject to client confidentiality where it applies.